

RDL-like sweep through inversion (msinv hull simulator)

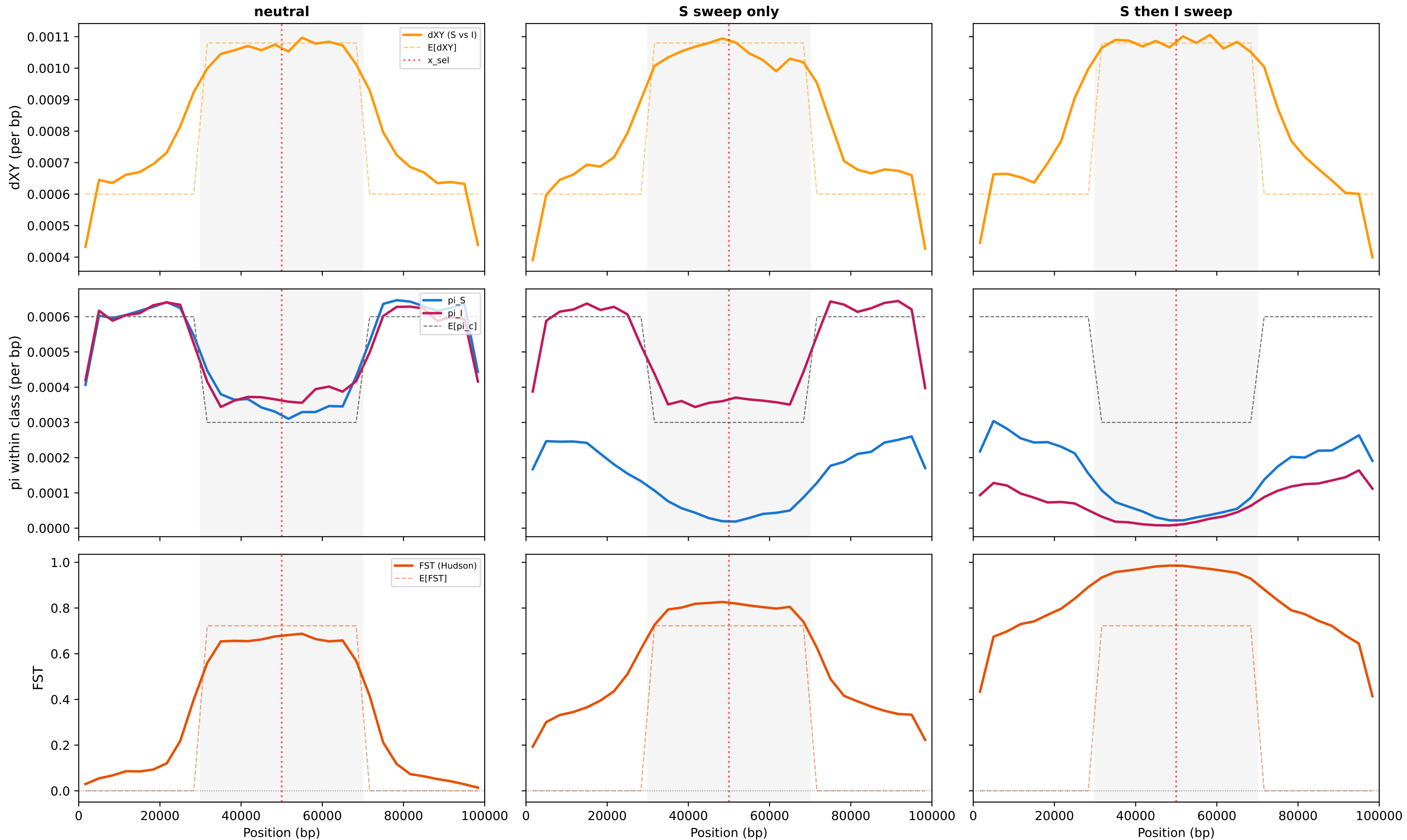


Figure. RDL (dieltrin resistance) sweep-through-inversion, modelled after Grau-Bové et al. (2020 MBE). Hitchhiking mode ($s=0.05$) with spatial decay $P(\text{linked}) = \exp(-r|x-x_{\text{sel}}|t_{\text{dur}})$. Three scenarios for $n_S=8 + n_I=8$ haplotypes. (Row 1) Cross-class dXY. (Row 2) Within-class π_S , π_I . (Row 3) Hudson FST. Dashed: structured coalescent predictions (no-flux baseline). Left: Neutral — π dips to $p^*\theta$ inside inv (isolated classes). Centre: S sweep only ($t=700$ gen) — π_S valley around x_{sel} , π_I unaffected. Right: S then I sweep ($t_S=700$, $t_I=300$ gen) — both collapse. Parameters: $N_e=50000$, $p_{\text{inv}}=0.5$, $t_{\text{inv}}=80000$ gen, $s=0.05$, $\gamma=1e-9$, $\mu=3e-9$, $r=1e-8$, $L=100$ kb, 100 reps. Command: `pixi run -e all python examples/empirical_rdl_sweep.py`